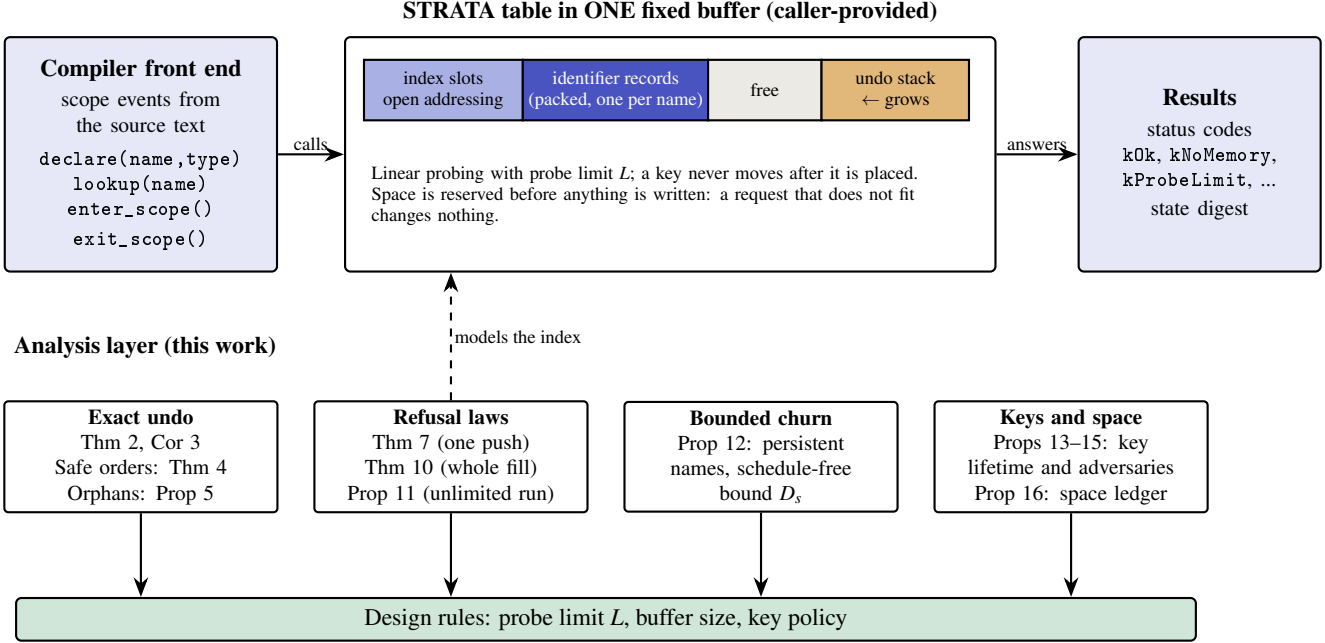

STRATA table in ONE fixed buffer (caller-provided)
Compiler front end
scope events from the source text
declare(name,type)
lookup(name)
enter_scope()
exit_scope()
calls
index slots open addressing
identifier records (packed, one per name)
free
undo stack ← grows
Linear probing with probe limit L; a key never moves after it is placed. Space is reserved before anything is written: a request that does not fit changes nothing.
answers
Results
status codes
kOk, kNoMemory, kProbeLimit, ...
state digest
models the index
Analysis layer (this work)
Exact undo
Thm 2, Cor 3
Safe orders: Thm 4
Orphans: Prop 5
Refusal laws
Thm 7 (one push)
Thm 10 (whole fill)
Prop 11 (unlimited run)
Bounded churn
Prop 12: persistent names, schedule-free bound D_s
Keys and space
Props 13–15: key lifetime and adversaries
Prop 16: space ledger
Design rules: probe limit L, buffer size, key policy